VRATTIKS

SOFTWARE REQUIREMENTS SPECIFICATION

Samaj Mobile Application

Member app for iOS and Android plus Admin Dashboard for community registration, matrimony, news, Panchang and directory

IEEE 830 / ISO/IEC/IEEE 29148 aligned

Document Metadata

VERSION	**1.0**	DATE	**2026-10-09**
AUTHOR	**Principal Software Architect, Vrattiks**	CLIENT	**Samaj Community (via Vrattiks Intelligence LLP)**
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TECH STACK	**React Native · FastAPI · SQLAlchemy · PostgreSQL**		

PURPOSE

This SRS converts the Samaj SOP (v1.1, 09/10/26) and the Samaj Mobile App PRD (v1.0) into verifiable functional and non-functional requirements. It is the baseline contract for design, development, testing and acceptance of the GA release, built on React Native, FastAPI with SQLAlchemy, and PostgreSQL.

Prepared by **Vrattiks — Software Architecture & Systems Engineering**

Internal · Confidential. This document is the contractual single source of truth for requirements between the client, development and QA.

Document Control

Revision History

Version	Date	Author	Description of Change
1.0	2026-10-09	Vrattiks Architecture	Initial SRS baseline from SOP v1.1 and PRD v1.0, including biodata consent & privacy flow.

Source Documents

Ref	Document	Version / Date	Notes
SOP	Samaj Mobile Application — Standard Operating Procedure	v1.1 · 09/10/26	Primary source for workflows, roles, error codes, security and release rules
PRD	Samaj Mobile Application — Product Requirements Document	v1.0 · 2026-10-09	User stories, priorities, NFR targets, assumptions and phase plan
Stack	Client-mandated technology stack	2026-10-09	React Native; FastAPI + SQLAlchemy; PostgreSQL

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the **Samaj Mobile Application**: a React Native member app for iOS and Android and a browser-based Admin Dashboard backed by a FastAPI/PostgreSQL service. It is written for the Samaj Super Admin and community leadership (scope sign-off), Vrattiks engineering (design and build), Vrattiks QA (test design) and auditors reviewing data protection.

1.2 Scope

Release 1.0 (GA, delivered through Alpha and Beta phases) covers OTP registration and admin verification, family additions, matrimony profiles with photo moderation and filtered search, consent-based biodata exchange, scheduled news and ads, push notifications, the daily Panchang with manual fallback, the member directory, city-scoped role-based administration with MFA, bulk CSV/XLSX import with deduplication, encrypted filtered export, audit logging, data retention and DPDP data-principal rights.

Out of scope: in-app chat or calling, payments or premium matrimony plans, Kundli/horoscope matching, a member-facing web app, AI match recommendations, event ticketing/RSVP, and automated KYC/Aadhaar verification.

0
Biodata exposures
To any third party, incl. admins

05:00
Panchang deadline
IST, 100 % of days

≤ 30 s
News push
95 % of devices

0
Import duplicates
Dedup on mobile number

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
Samaj	The community the application serves.
Biodata	Detailed personal and family profile used for matrimony; shared only between two members after an accepted request.
Public summary	Matrimony fields visible in search: photo, age, education, city, family background category.
Panchang	Hindu almanac; in scope are the daily Tithi (lunar day), Nakshatra (lunar mansion) and Yoga.
City Admin	The single administrator appointed per city by the Super Admin; performs the SOP's Moderator and Content Admin duties (ASM-02).
Third party	Any person other than the two members of an accepted biodata request — including other members, City Admins and the Super Admin.
OTP / TOTP	One-time password sent by SMS / time-based one-time password from an authenticator app (admin MFA).

Term	Definition
RBAC	Role-based access control.
FCM / APNs	Firebase Cloud Messaging / Apple Push Notification service.
DLT	Distributed Ledger Technology registration of SMS templates mandated by TRAI in India.
DPDP Act	Digital Personal Data Protection Act, 2023 (India).
ORM	Object-relational mapper (SQLAlchemy).
RPO / RTO	Recovery point objective / recovery time objective.
MoSCoW	Must / Should / Could / Won't prioritisation.

1.5 Document Overview

Section 2 describes the product context and constraints; Section 3 the external interfaces; Section 4 the functional requirements by system feature; Section 5 the non-functional requirements; Section 6 the data requirements; Section 7 traces every SOP step to requirements and tests; Section 8 lists open questions and risks. Each requirement's source cites the SOP section and, where applicable, the PRD requirement ID.

2. Overall Description

2.1 Product Perspective

Samaj is a new, self-contained system replacing offline registers, printed Panchang calendars, messaging groups and spreadsheets. It consists of three components: the **member app** (React Native, iOS and Android), the **Admin Dashboard** (web, desktop browsers) and the **backend API** (FastAPI with SQLAlchemy over PostgreSQL). It integrates with an SMS OTP gateway, FCM/APNs for push, an external astrology API, object storage for media and an email service for admin alerts.

System context

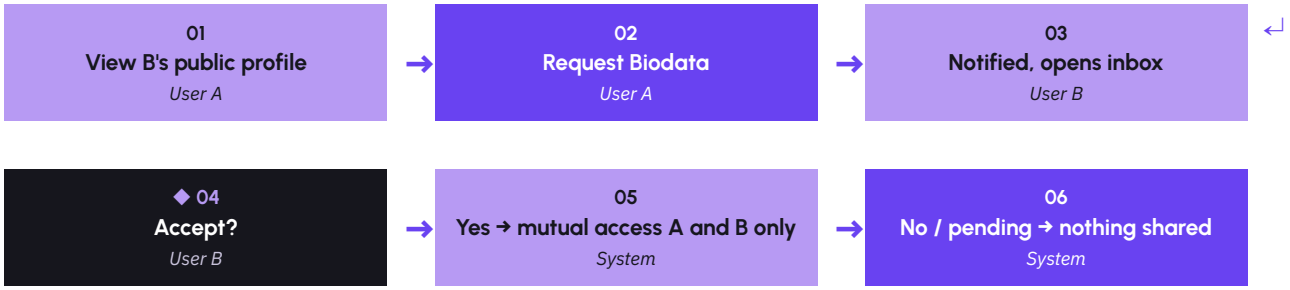
Component	Responsibility	Technology
Member app	Registration, profile, matrimony, biodata requests, news, Panchang, directory	React Native (iOS 15+, Android 9+)
Admin Dashboard	Approvals, content, Panchang override, import/export, audit, settings	Web SPA — framework to be confirmed (OQ-03)
Backend API	Business rules, authorisation, scheduling, integrations	Python FastAPI, SQLAlchemy 2.x ORM, Alembic migrations
Database	System of record for all entities and audit log	PostgreSQL 15+
Background workers	Scheduled publishing, Panchang fetch, imports, retention jobs, push fan-out	Task queue/scheduler — to be confirmed (OQ-11)
Object storage	Photos, ID documents, ad creatives (encrypted)	S3 or equivalent (OQ-06)

2.2 Product Functions

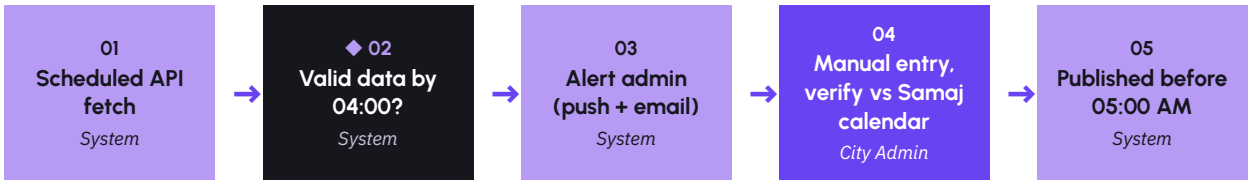
Member onboarding — happy path (SOP §6.1)



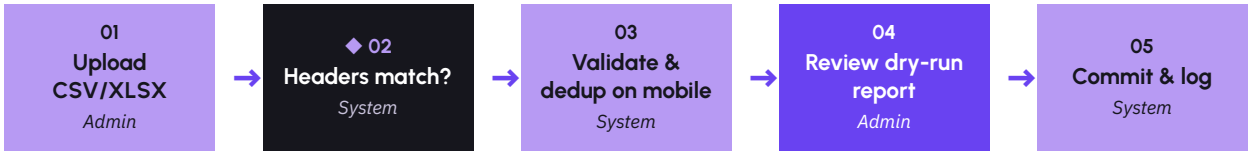
Biodata exchange (SOP §6.2 Steps 1–4)



Daily Panchang with fallback (SOP §6.4, ERR_API_PN)



Weekly member import (SOP §5.4, §6.5)



2.3 User Classes and Characteristics

User Class	Description	Frequency	Technical Proficiency
Super Admin	One per deployment. Assigns City Admins, governs security, reviews audit logs monthly, manages Layout Settings.	Daily	High
City Admin	One per city. Approves members, family additions and matrimony photos; schedules news/ads; Panchang override; import/export.	Daily	Medium
Member — Matrimony seeker	22–35 years, or a parent acting for them. Creates matrimony profile, searches, exchanges biodata.	Weekly	Medium
Member — Community member	30–70 years, often head of household. Panchang, news, directory, family additions; needs large text.	Daily	Low
IT Support / Senior Technical Staff	Monitoring, backups, deployments, DB access over VPN.	As needed	High

2.4 Operating Environment

- Member app: Android 9+ and iOS 15+, phones and tablets (small, medium, large screens), on 4G/Wi-Fi.
- Admin Dashboard: latest two versions of Google Chrome and Safari on desktops/laptops with ≥ 8 GB RAM (SOP §4).
- Backend: containerised FastAPI services (ASGI, Uvicorn/Gunicorn) behind a TLS-terminating load balancer, PostgreSQL 15+ managed instance, in an India cloud region (AWS or Azure — OQ-06).
- Environments: Development, Staging (mirrors production; SOP §9 checklist) and Production.

2.5 Design and Implementation Constraints

CONSTRAINT · **Client-mandated constraints**

- **CON-01** Member app shall be built in React Native from a single codebase for iOS and Android.
- **CON-02** Backend shall be built with Python FastAPI, using SQLAlchemy as the ORM; schema changes via versioned migrations.
- **CON-03** PostgreSQL shall be the system of record for members, matrimony, biodata, content, Panchang and audit logs.
- **CON-04** Admin Dashboard access requires MFA and RBAC; production DB access only over VPN by senior technical staff (SOP §4).
- **CON-05** Releases follow SOP §10: 48 h code freeze, staging sign-off, full backup, CI/CD deploy at 02:00 AM IST, smoke test.
- **CON-06** Media limits: images ≤ 2 MB; full-screen ads 1080×1920 (SOP §6.3, §7).
- **CON-07** Push notifications via Firebase Cloud Messaging (SOP §12), with APNs for iOS.

2.6 Assumptions and Dependencies

ASSUMPTIONS · **Assumptions (to be validated)**

- **ASM-01** Launch scale ≈ 10,000 members across ≈ 10 cities, growing to 50,000 within 2 years.
- **ASM-02** SOP 'Moderator' and 'Content Admin' duties belong to the City Admin role (SOP §3 defines three roles).
- **ASM-03** 'Third party' in the biodata privacy rule includes City Admins and the Super Admin; admins never see biodata.
- **ASM-04** Mobile number is the unique member identifier for deduplication.
- **ASM-05** Panchang alert cut-off is 04:00 AM IST, leaving one hour for manual entry; one Panchang per deployment (single reference location).
- **ASM-06** Admin Dashboard is a desktop web application.
- **ASM-07** Pending members may view Panchang and News only.
- **ASM-08** A declined biodata request cannot be re-sent for 30 days.
- **ASM-09** Daily cap of 20 biodata requests per member (anti-spam).
- **ASM-10** Edits to verified identity fields (name, DOB, mobile) return the profile to City Admin review.

DEPENDENCIES · **External dependencies**

- **DEP-01** Astrology/Panchang API provider and licence (OQ-04).
- **DEP-02** SMS OTP gateway with TRAI DLT-registered templates.
- **DEP-03** Firebase project (FCM) and Apple Developer account (APNs).
- **DEP-04** Cloud host, object storage, VPN, CI/CD and monitoring stack (OQ-06).
- **DEP-05** Google Play and Apple App Store developer accounts by Beta.
- **DEP-06** Written Samaj bylaws and photo guidelines for moderation.

3. External Interface Requirements

Interfaces between the system and its users, devices, external software and networks.

ID	Interface	Type	Requirement
UIR-01	Member app — Login/OTP	User	Phone input with +91 default, 6 OTP boxes with auto-fill, resend timer (30 s), clear error states.
UIR-02	Member app — Home	User	Today's Panchang card, latest news, ad slot, quick links to Matrimony and Directory.
UIR-03	Member app — Matrimony	User	Filter sheet (age, education, location, family background); profile cards with public summary; 'Request Biodata' button showing Request / Pending / View Biodata state.
UIR-04	Member app — Biodata requests	User	Received / Sent tabs with Accept / Decline actions and status badges.
UIR-05	Member app — Accessibility	User	Large-text mode, dynamic type to 200 %, touch targets ≥ 44×44 pt, light and dark themes; languages per OQ-05.
UIR-06	Admin Dashboard	User	Approvals (tabbed queues), Content Management with scheduler and calendar, Panchang status & manual form, Import/Export, Admin & Audit, Layout Settings, Archive Recovery.
UIR-07	Error presentation	User	Admin-facing failures show SOP code (ERR_AUTH_01, ERR_DB_CON, ERR_IMG_UL, ERR_API_PN) with corrective action.
HWR-01	Mobile device	Hardware	Camera and photo library access for profile/ID images (permission prompts); push notification permission.
HWR-02	Admin workstation	Hardware	Desktop/laptop ≥ 8 GB RAM, stable internet; authenticator app on a phone for TOTP.
SWR-01	Backend REST API	Software	JSON over HTTPS, versioned under /api/v1, OpenAPI 3 spec generated by FastAPI; JWT bearer auth.
SWR-02	Astrology / Panchang API	Software	HTTPS REST pull returning Tithi, Nakshatra, Yoga per date and location; timeout 10 s, 3 retries with back-off. Provider to be confirmed (DEP-01).
SWR-03	SMS OTP gateway	Software	HTTPS API, DLT template IDs, delivery-status callback; secondary provider for failover.
SWR-04	FCM / APNs	Software	FCM HTTP v1 API for Android and iOS (APNs via FCM); token registration and invalidation handling.
SWR-05	Object storage	Software	S3-compatible API; private buckets; server-side AES-256 encryption; time-limited pre-signed URLs (≤ 15 min).
SWR-06	Email service	Software	SMTP/API for admin alerts (Panchang failure, escalations, import reports).

ID	Interface	Type	Requirement
SWR-07	PostgreSQL	Software	Accessed only by the backend via SQLAlchemy over TLS; no direct client access.
CIR-01	Transport security	Communications	TLS 1.2+ on all external and internal connections; HSTS on dashboard; certificate pinning in the mobile app — to be confirmed.
CIR-02	Admin DB access	Communications	Production database reachable only through VPN by senior technical staff.

4. Functional Requirements

Requirements are grouped into 16 system features and numbered continuously FR-01 to FR-100. Priorities use MoSCoW for the GA release; the source cites the SOP section and the originating PRD requirement.

4.1 SF-01 — Mobile Registration & OTP Login

PRIORITY Must | ACTORS Member, SMS OTP Gateway

Members join and sign in with a mobile number and a one-time password on iOS and Android; there are no passwords. Registration captures the member's profile and DPDP consent and places the account in the approval queue.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member enters a mobile number (+91 default) and taps Continue	System checks the number against existing members; if new, sends a 6-digit OTP by SMS
2	Member enters the OTP	System validates the code (expiry, attempts) and opens the profile form
3	Member submits profile and consent	System stores the profile with status Pending Approval and routes it to the City Admin of the selected city
4	Existing member enters a registered number	System informs the member the number is registered and switches to the login flow

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	The system shall allow a member to register using a mobile number and a 6-digit SMS OTP, without a password, on both iOS and Android apps.	Must	Registration with a valid OTP completes on an Android 9+ and an iOS 15+ device in staging.
FR-02	The system shall allow a registered member to log in using mobile number and SMS OTP.	Must	A registered Active member receives an OTP and reaches the Home screen after entering it.
FR-03	The system shall reject an OTP that is older than 5 minutes or after 3 failed attempts, and require a new OTP.	Must	An OTP entered at 5 min 01 s, or a 4th attempt, returns an 'OTP expired/locked' error.
FR-04	The system shall limit OTP sends to 5 per mobile number per rolling hour.	Must	The 6th OTP request within 60 minutes returns HTTP 429 and no SMS is sent.

ID	Requirement	Priority	Acceptance Criteria
FR-05	The system shall check every new registration for an existing member with the same mobile number and block duplicate registration.	Must	Registering a number already in the members table returns 'Number already registered' and offers Login; no new record is created.
FR-06	The system shall capture the member's name, gender, date of birth, city, address and household details on the profile form, validating mandatory fields before submission.	Must	Submitting with any mandatory field empty shows a field-level error and no record is saved.
FR-07	The system shall record the member's explicit, purpose-specific consent (DPDP Act 2023) with timestamp and notice version before creating the profile.	Must	Profile submission is disabled until the consent checkbox is ticked; the consent row stores member ID, notice version and timestamp.
FR-08	The system shall set a newly submitted profile to status 'Pending Approval' and restrict the member to Panchang and News until approved.	Must	A pending member sees Panchang and News; Matrimony and Directory show 'Awaiting approval'.

4.2 SF-02 — Member Verification & Approval

PRIORITY Must | ACTORS City Admin, Member, Push Service

City Admins verify new members of their city, optionally against an uploaded identity document, and approve or reject them. The member is notified of the outcome.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member uploads an optional ID document	System validates format and size and stores it encrypted, visible only to the member's City Admin and the Super Admin
2	City Admin opens the Approvals queue	System lists pending registrations for that admin's city only, oldest first
3	City Admin clicks Approve or Reject (with reason)	System updates status, writes an audit entry and notifies the member
4	Registration pending > 72 h	System escalates to the Super Admin

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-09	The system shall allow a member to upload an optional identity document in JPG, PNG or PDF format of at most 2 MB.	Should	A 2.1 MB file or a .gif is rejected with ERR_IMG_UL; a 1.9 MB PDF is accepted.

ID	Requirement	Priority	Acceptance Criteria
FR-10	The system shall restrict viewing of identity documents to the member's City Admin and the Super Admin.	Must	A request for the document by any other user returns HTTP 403 and is logged.
FR-11	The system shall present each City Admin with a queue of pending registrations limited to that admin's assigned city.	Must	A City Admin of City X cannot list or open pending members of City Y (API returns 403).
FR-12	The system shall allow a City Admin to approve a pending member, changing the status to Active.	Must	After Approve, member status = Active and Matrimony/Directory unlock on next app refresh.
FR-13	The system shall require a reason when a City Admin rejects a registration and show that reason to the member.	Must	Reject is disabled until a reason is entered; the member sees the reason in-app.
FR-14	The system shall send the member a push notification on approval or rejection within 30 seconds, falling back to SMS if no valid push token exists.	Should	With a valid FCM/APNs token the push arrives ≤ 30 s; with no token an SMS is sent.
FR-15	The system may allow a City Admin to approve multiple selected pending members in one action.	Could	Selecting 10 members and clicking Bulk Approve activates all 10 and writes 10 audit entries.
FR-16	The system shall notify the Super Admin when any registration remains pending for more than 72 hours.	Should	A registration aged 72 h 01 min appears in the Super Admin escalation list and triggers an email.

4.3 SF-03 — Family Addition

PRIORITY Should ACTORS Member, City Admin

Active members request that relatives be added to their household. Additions appear in the directory only after City Admin approval.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member submits a family-addition request	System creates a pending family record linked to the member's household and queues it for the City Admin
2	City Admin approves or rejects	System publishes the relative under the household in the directory, or notifies the member with the reason

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-17	The system shall allow an Active member to request the addition of a family member to their household, capturing name, relationship, gender and date of birth.	Should	A submitted request appears in the City Admin's Family Additions queue with status Pending.
FR-18	The system shall display an added family member in the directory only after City Admin approval.	Should	A pending relative is absent from directory search; after approval it appears under the household.

4.4 SF-04 — Matrimony Profile & Photo Moderation

PRIORITY Must ACTORS Member (or family member acting for them), City Admin

Active members create a matrimony profile split into a public summary (photo, age, education, city) and a private biodata section. Photos are moderated before the profile becomes searchable.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member creates a matrimony profile with photos	System saves it as Pending Moderation and hides it from search
2	City Admin reviews photos against community guidelines	System marks the profile Approved (searchable) or Rejected (member told why)
3	Member edits photos on an approved profile	System returns the new photos to moderation; previously approved photos remain visible

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-19	The system shall allow only Active members to create a matrimony profile.	Must	A Pending member tapping 'Create Matrimony Profile' receives 'Available after approval'.
FR-20	The system shall store each matrimony profile as two separately access-controlled parts: a public summary and a private biodata section.	Must	Schema review confirms biodata in a separate table/columns governed by the biodata pair-access check (FR-35).
FR-21	The system shall allow a member to indicate that the matrimony profile is managed by a family member acting on the candidate's behalf.	Should	Profile shows 'Managed by: Parent/Sibling' when the flag is set.

ID	Requirement	Priority	Acceptance Criteria
FR-22	The system shall keep a new matrimony profile out of search results until its photos are approved by the City Admin.	Must	A pending profile is not returned by any search query.
FR-23	The system shall allow the City Admin to reject a matrimony photo with a reason that is shown to the member.	Must	Rejected photo is removed from the profile and the member sees the reason.
FR-24	The system shall accept matrimony photos only in JPG or PNG format of at most 2 MB each.	Must	Uploading a 3 MB image returns ERR_IMG_UL with the corrective message.

4.5 SF-05 — Matrimony Search & Filters

PRIORITY Must SOURCE SOP §6.2, §9 · PRD FR-10–11 ACTORS Member

Active members browse approved matrimony profiles, filtering by age, education, location and family background. Results show public summaries only.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member opens Matrimony search and sets filters	System returns approved profiles matching all filters, public summary fields only
2	Profile approved or edited	System reflects the change in search within the index refresh interval

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-25	The system shall allow members to filter matrimony profiles by age range, education, location and family background, in any combination.	Must	For a seeded data set, each filter combination returns exactly the expected profile IDs.
FR-26	The system shall return only public-summary fields in matrimony search results and never biodata fields.	Must	Search API response schema contains no biodata fields (contract test).
FR-27	The system shall exclude the searching member's own profile and any Pending, Rejected or Archived profiles from results.	Should	Seeded Pending/Archived profiles never appear in results.
FR-28	The system shall refresh the matrimony search index at intervals of no more than 4 hours.	Should	A profile approved at T appears in search results by T + 4 h.

4.6 SF-06 — Biodata Request & Consent-Based Exchange

PRIORITY Must ACTORS User A (requester), User B (recipient), Push Service

Biodata is exchanged only with consent. User A requests User B's biodata; only when User B accepts do A and B gain mutual access to each other's biodata. Biodata is never visible to any third party — other members or administrators.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	User A taps 'Request Biodata' on User B's profile	System creates a Pending request, notifies User B and shows 'Pending' to A; no biodata is shared
2	User B opens the request inbox and taps Accept	System grants mutual access: A and B can each view the other's biodata
3	User B taps Decline	System shares nothing, shows 'Declined' to A and locks resend for 30 days
4	User C, or any admin, requests A's or B's biodata	System returns HTTP 403 / omits fields and writes a security log entry
5	Either party withdraws an accepted exchange	System revokes access for both immediately

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-29	The system shall allow an Active member (User A) to send a biodata request to another member (User B) from User B's matrimony profile.	Must	Tapping 'Request Biodata' creates a request with status Pending and the button changes to 'Pending'.
FR-30	The system shall allow at most one open (Pending or Accepted) biodata request per member pair and reject duplicates.	Must	A second request from A to B, or from B to A, while one is open returns 'Request already exists'.
FR-31	The system shall notify User B of a new biodata request by push notification.	Should	User B's device receives the push ≤ 30 s after A sends the request.
FR-32	The system shall allow User B to Accept or Decline a pending biodata request.	Must	Both actions are available on the Received tab and update the request status.
FR-33	The system shall, on acceptance, grant User A and User B access to each other's biodata.	Must	After Accept, A can open B's biodata and B can open A's biodata.
FR-34	The system shall share no biodata while a request is Pending or after it is Declined.	Must	Biodata endpoint returns 403 for both A and B for Pending and Declined requests.

ID	Requirement	Priority	Acceptance Criteria
FR-35	The system shall enforce the biodata pair-access check server-side on every request that could return biodata fields, returning biodata only to the two members of an accepted request.	Must	Automated privacy suite: user C, City Admin and Super Admin calls to biodata, profile, search, directory and export endpoints never return biodata fields.
FR-36	The system shall exclude biodata from all administrator views, including the Admin Dashboard, approval screens and exports.	Must	Dashboard profile view shows the public summary only; DOM/API inspection shows no biodata fields.
FR-37	The system shall log every denied attempt to access biodata with actor, target, endpoint and timestamp.	Must	A denied call produces one security-log row with all four attributes.
FR-38	The system shall notify User A when User B accepts or declines a request.	Should	User A receives a push and the Sent tab shows the new status.
FR-39	The system shall block User A from re-sending a request to User B for 30 days after a decline.	Should	Request attempt on day 29 is rejected; on day 31 it succeeds.
FR-40	The system shall show each member Sent and Received request tabs with Pending, Accepted and Declined status.	Should	Tabs list all requests with correct status badges.
FR-41	The system may allow either party to withdraw an accepted exchange, revoking biodata access for both immediately.	Could	After withdrawal, the next biodata call by either party returns 403.
FR-42	The system may limit each member to 20 new biodata requests per day.	Could	The 21st request in a calendar day (IST) returns a daily-limit message.

SECURITY · **Biodata privacy rule (SOP §6.2, §8)**

Shared biodata is visible only to User A and User B after User B accepts. It must never be shown to any third party — other members, City Admins or the Super Admin — through any screen, search, directory, export, log or API response. The pair-access check is enforced in the backend data-access layer (not the client), biodata fields are encrypted at field level, and a dedicated automated privacy test suite is a release gate.

4.7 SF-07 — News & Ads Content Management

PRIORITY Must | ACTORS City Admin, Super Admin, Member

Admins create news items and ad campaigns in the Content Management tab, upload media, set visibility windows and schedule them. Content goes live and expires automatically; members see it in a reverse-chronological feed.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Admin selects 'Add New' (News) or 'New Campaign' (Ads)	System opens the content form with media uploader and scheduler
2	Admin uploads media	System validates format, size and, for full-screen ads, 1080×1920 dimensions
3	Admin sets start/end date-time and target city, clicks Schedule	System queues the item with status Scheduled
4	Start time reached	System sets the item Live, shows it in the feed and triggers notifications
5	End time reached	System sets the item Expired and removes it from the feed

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-43	The system shall allow an admin to create a news item with title, body, optional image, start date-time, end date-time and target audience.	Must	A saved news item persists all six fields.
FR-44	The system shall allow an admin to create an ad campaign with a banner or full-screen creative, start date and end date.	Should	A saved campaign persists creative type, media and dates.
FR-45	The system shall reject a full-screen ad creative whose dimensions are not 1080×1920 pixels.	Should	A 1080×1080 image uploaded as full-screen returns a dimension error.
FR-46	The system shall reject an end date-time that is earlier than or equal to the start date-time.	Must	Saving with end ≤ start shows a validation error.
FR-47	The system shall publish scheduled content automatically at its start time and expire it automatically at its end time.	Must	Item scheduled for 10:00 is visible at 10:01 and absent 1 min after its end time.
FR-48	The system shall restrict City Admins to publishing content for their own city and allow the Super Admin to publish to all cities.	Should	A City Admin's audience selector lists only their city; API rejects other city IDs with 403.
FR-49	The system shall display live news to members in reverse-chronological order, filtered to their city plus all-city items.	Must	Feed order matches publish time descending; another city's items are absent.
FR-50	The system shall place ads in the feed according to the placement rules defined in Layout Settings.	Should	With 'ad every 5 items', the 6th card is an ad.

ID	Requirement	Priority	Acceptance Criteria
FR-51	The system may allow the Super Admin to change standard ad dimensions and placements in a Layout Settings module.	Could	Changing placement in Layout Settings is reflected in the app feed on next refresh.
FR-52	The system may provide a calendar view of scheduled, live and expired content.	Could	Calendar displays each item on its active dates with status colour.

4.8 SF-08 — Push Notifications

PRIORITY Should SOURCE SOP §6.1 Step 4, §9, §12 · PRD FR-06, FR-20 ACTORS System, Member, FCM / APNs

The system registers device tokens and sends push notifications for account activation, news going live, biodata requests and Panchang alerts.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	App launches and permission is granted	System registers or refreshes the device's FCM token against the member
2	News item goes live	System sends a push to the target audience
3	Token reported invalid by FCM	System marks the token stale and stops sending to it

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-53	The system shall register and refresh the device push token for each logged-in member.	Must	After login, the member's token row exists with last-seen timestamp.
FR-54	The system shall send a push notification to the target audience when a news item goes live.	Should	≥ 95 % of test devices receive the push ≤ 30 s after go-live.
FR-55	The system shall mark a device token as invalid when the push provider reports it as unregistered and stop sending to it.	Should	After an 'UNREGISTERED' response, no further sends target that token.
FR-56	The system shall allow members to opt out of news and ads notifications while keeping account and biodata notifications.	Should	With news notifications off, a news go-live sends nothing to that member; biodata pushes still arrive.

4.9 SF-09 — Daily Panchang

PRIORITY Must SOURCE SOP §2, §6.4, §7 ERR_API_PN · PRD FR-23–25 ACTORS System, Astrology API, City Admin / Content Admin, Member

The system fetches the day's Tithi, Nakshatra and Yoga from an external astrology API and shows it in the app. If the API fails, the responsible admin is alerted and enters the values manually before 05:00 AM IST, verified against the Samaj Panchang calendar.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Scheduled job runs (evening pre-fetch and early-morning retry)	System calls the astrology API and stores the next day's values with source = API
2	API times out or returns invalid data by 04:00 AM IST	System raises ERR_API_PN and sends push + email to the responsible admin
3	Admin enters Tithi, Nakshatra and Yoga manually	System stores values with source = Manual and editor ID; manual overrides API values
4	Member opens the Panchang screen	System shows the day's values with date

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-57	The system shall retrieve the daily Tithi, Nakshatra and Yoga from the configured external astrology API.	Must	Daily job stores a record per date with three values and source = API.
FR-58	The system shall pre-fetch the next day's Panchang the evening before and retry until valid data is stored.	Should	Records for D+1 exist by 22:00 IST on day D in normal operation.
FR-59	The system shall display the current day's Tithi, Nakshatra and Yoga with the date on the Home and Panchang screens.	Must	Opening the app after 05:00 AM IST shows today's values.
FR-60	The system shall raise error ERR_API_PN and alert the responsible admin by push and email if no valid Panchang exists for the day by 04:00 AM IST.	Should	Simulated API outage produces an alert at 04:00 with code ERR_API_PN.
FR-61	The system shall provide an admin form to enter or correct the day's Tithi, Nakshatra and Yoga manually.	Must	Form saves all three values for a selected date.
FR-62	The system shall give manually entered Panchang values precedence over API values for the same date and record source, editor and timestamp.	Must	After manual entry, app shows manual values; edit history shows editor and time.
FR-63	The system shall require the admin to confirm that manual values were verified against the Samaj Panchang calendar before saving.	Should	Save is disabled until the verification checkbox is ticked.
FR-64	The system may allow members to view the Panchang for other dates using a date switcher.	Could	Selecting tomorrow shows D+1 values if available.

4.10 SF-10 — Member Directory

PRIORITY Must SOURCE SOP §2, §3 · PRD FR-26–27 ACTORS Member

Active members search the community directory by name, city and family. Results show approved directory fields only and never biodata.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member searches by name, city or family	System returns Active members and approved family members with directory fields
2	Member opens a household	System shows the household and its approved family members
3	Member hides phone/address	System omits those fields from other members' results

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-65	The system shall allow Active members to search the directory by name, city and family/household.	Must	Search 'Patel' in City X returns only Active matching members of City X when that filter is set.
FR-66	The system shall return only approved directory fields in directory results and never biodata fields.	Must	Directory API contract contains no biodata fields.
FR-67	The system shall display a household view listing the member and approved family members.	Should	Household screen lists all approved relatives.
FR-68	The system may allow a member to hide their phone number and address from the directory.	Could	With 'hide phone' on, other members' results show the phone as hidden.

4.11 SF-11 — Role-Based Access & Admin Authentication

PRIORITY Must SOURCE SOP §3, §4, §7 ERR_AUTH_01 · PRD FR-28–29 ACTORS Super Admin, City Admin

Three roles — Super Admin, City Admin and Member — with city-scoped data access. All Admin Dashboard logins require multi-factor authentication.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Super Admin assigns a City Admin to a city	System enforces one City Admin per city and records the assignment
2	Admin logs in to the dashboard	System requires credentials plus a TOTP second factor

#	User / System Stimulus	System Response
3	MFA verification fails (ERR_AUTH_01)	System shows the code with corrective action and allows session reset

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-69	The system shall implement three roles — Super Admin, City Admin and Member — and authorise every API request against the caller's role server-side.	Must	Role-permission matrix tests pass for every endpoint; client-side hiding alone is not accepted.
FR-70	The system shall allow the Super Admin to assign exactly one City Admin per city and reject a second assignment to the same city.	Must	Assigning a second admin to City X returns 'City already has an admin'.
FR-71	The system shall limit a City Admin's access to members, content, imports and exports of their assigned city.	Must	Cross-city requests by a City Admin return 403.
FR-72	The system shall require multi-factor authentication (password plus TOTP) on every Admin Dashboard login.	Must	Login without a valid TOTP is refused; 100 % of admin sessions carry an MFA claim.
FR-73	The system shall terminate an admin session after 30 minutes of inactivity.	Must	An idle session at 31 min requires re-authentication.
FR-74	The system shall allow the Super Admin to reset an admin's session and MFA enrolment when ERR_AUTH_01 occurs.	Should	After reset, the admin can re-enrol TOTP and log in.

> SECURITY · **Admin access**
> MFA is mandatory for every Admin Dashboard login. City scope is enforced server-side on every query. Database access is restricted to senior technical staff over VPN and all sessions are logged.

4.12 SF-12 — Bulk Member Import & Deduplication

PRIORITY Must SOURCE SOP §5.4, §6.5, §9 · PRD FR-30–31, FR-38 ACTORS City Admin, Super Admin

Admins upload weekly member lists as CSV or XLSX. Headers are validated strictly against the published template; rows are deduplicated against existing members and a row-level report is produced before commit.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Admin downloads the template	System provides the current CSV/XLSX template

#	User / System Stimulus	System Response
2	Admin uploads a file	System validates headers; on mismatch the whole file is rejected before any write
3	Headers valid	System validates rows, matches on phone number (secondary: name + DOB) and shows a dry-run report: new, update, conflict, error
4	Admin confirms commit	System inserts new rows, updates matches, leaves conflicts for review, logs the import

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-75	The system shall accept member import files in CSV and XLSX format.	Must	Both formats import successfully; other extensions are rejected.
FR-76	The system shall reject an entire import file whose column headers do not exactly match the published template, before writing any data.	Must	A file with one renamed header is rejected and no rows change.
FR-77	The system shall match imported rows to existing members on mobile number and update the existing record instead of creating a duplicate.	Must	Importing a file containing 100 existing numbers creates 0 new members for those rows.
FR-78	The system shall flag as a conflict any row whose mobile number matches an existing member but whose name and date of birth differ, and exclude it from automatic update.	Must	Conflict rows appear in the review list and the existing record is unchanged.
FR-79	The system shall present a dry-run report of new, updated, conflicting and invalid rows, with row-level error messages, before commit.	Must	Report counts equal the seeded file's expected counts.
FR-80	The system shall process an import of 10,000 rows, including validation and deduplication, within 2 minutes.	Should	Timed test with a 10,000-row file completes ≤ 120 s.
FR-81	The system may run a recurring weekly import from a designated upload location.	Could	A file placed in the location is processed at the scheduled time and a report emailed.

4.13 SF-13 — Encrypted Filtered Export

PRIORITY Must SOURCE SOP §6.5, §8 · PRD FR-32 ACTORS City Admin, Super Admin

Admins export filtered sub-groups of members for community events. Every export is password-encrypted and excludes biodata.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Admin sets filters and clicks 'Export Filtered'	System prompts for a password
2	Admin enters password	System generates an AES-256 encrypted file, logs the export and offers download

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-82	The system shall allow admins to export members matching selected filters (e.g. age band, area, city) via 'Export Filtered'.	Must	Exported rows equal the filtered result count.
FR-83	The system shall encrypt every export with an admin-supplied password (minimum 8 characters) using AES-256 before download.	Must	The file cannot be opened without the password; password of 7 chars is refused.
FR-84	The system shall exclude all biodata fields from every export.	Must	Export column list contains no biodata fields.
FR-85	The system shall not store or log the export password.	Must	Code review and log inspection show no password persistence.

4.14 SF-14 — Audit Logging, Monitoring & Error Codes

PRIORITY Must SOURCE SOP §5.5, §7, §8, §11 · PRD FR-33, FR-36 ACTORS System, Super Admin, IT Support

Every admin action is written to an append-only audit log reviewed monthly by the Super Admin. Operational errors surface with SOP error codes and corrective actions; logs and feedback are reviewed daily.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Admin performs approve/reject/edit/import/export/override	System writes an audit entry (actor, action, target, before/after, IP, timestamp)
2	Super Admin opens the audit view	System shows filterable entries with CSV download
3	Known failure occurs	System displays the SOP error code and corrective action to the admin and raises a monitoring alert where applicable

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-86	The system shall record every admin action (approve, reject, edit, import, export, Panchang override, role assignment) in an audit log with actor, action, target, timestamp and source IP.	Must	Each listed action produces exactly one audit row with all fields.
FR-87	The system shall make audit log entries append-only so that no user, including the Super Admin, can edit or delete them.	Must	UPDATE/DELETE on the audit table are refused for the application role.
FR-88	The system shall provide the Super Admin an audit view filterable by date range, admin, action and city, with CSV download.	Must	Filters return expected rows; CSV matches the screen.
FR-89	The system shall display the SOP error codes ERR_AUTH_01, ERR_DB_CON, ERR_IMG_UL and ERR_API_PN with their corrective actions on the Admin Dashboard.	Should	Each simulated failure shows its code and SOP corrective text.
FR-90	The system shall raise a real-time monitoring alert on ERR_DB_CON and when the API error rate exceeds 2 % over 5 minutes.	Should	Fault injection triggers an alert to the on-call channel within 2 minutes.
FR-91	The system may allow members to submit feedback or report a profile, and show submissions in an Admin Dashboard queue for daily review.	Could	A submitted report appears in the queue with reporter, target and text.

4.15 SF-15 — Data Retention, Archive & Recovery

PRIORITY Must SOURCE SOP §8, §12 · PRD FR-34–35 ACTORS System, Member, City Admin, Super Admin

Inactive profiles are archived after 12 months and purged after 24 months. Deleted profiles can be recovered within 30 days using Archive Recovery.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member inactive for 11 months	System notifies the member that the profile will be archived in 30 days
2	Member inactive for 12 months	System archives the profile and hides it from search and directory
3	Profile inactive for 24 months	System permanently purges personal data
4	Admin uses Archive Recovery within 30 days of deletion	System restores the profile and logs the action

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-92	The system shall notify a member 30 days before their profile is archived for inactivity.	Should	A member inactive for 11 months receives push/SMS notice.
FR-93	The system shall archive a member profile automatically after 12 months without login.	Must	A profile with last login 12 months + 1 day ago has status Archived and is absent from search/directory.
FR-94	The system shall purge personal data of a profile 24 months after the last login, retaining only anonymised aggregates and audit entries.	Must	Purge job removes PII columns, biodata and media for qualifying profiles.
FR-95	The system shall restore an archived profile to Active when the member logs in again before purge.	Should	Login by an archived member reactivates the profile (re-approval per ASM-10).
FR-96	The system shall allow an admin to recover a deleted member profile via 'Archive Recovery' within 30 days of deletion.	Should	Profile deleted 29 days ago is recoverable; at 31 days the option is unavailable.

4.16 SF-16 — Data Principal Rights (DPDP Act 2023)

PRIORITY Must SOURCE SOP §8 · PRD NFR-13 ACTORS Member, Super Admin

Members can view, correct and erase their personal data, withdraw consent, and reach a grievance contact, as required by India's Digital Personal Data Protection Act 2023.

Stimulus / Response Sequence

#	User / System Stimulus	System Response
1	Member requests a copy of their data	System generates a summary of personal data held
2	Member requests account deletion	System queues an erasure request, confirms by OTP and completes it within 30 days
3	Member raises a grievance	System records it and routes it to the Super Admin / grievance officer

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-97	The system shall allow a member to view and correct their personal data in-app.	Must	Edited fields persist; corrections to verified fields return the profile to review where ASM-10 applies.

ID	Requirement	Priority	Acceptance Criteria
FR-98	The system shall allow a member to request erasure of their account, confirmed by OTP, and complete the erasure within 30 days.	Must	Erasure request status reaches Completed ≤ 30 days; PII removed.
FR-99	The system shall allow a member to withdraw consent, which shall trigger the erasure workflow.	Must	Withdrawing consent creates an erasure request.
FR-100	The system shall display grievance officer contact details and allow a member to submit a grievance in-app.	Should	Grievance form submission appears in the Super Admin queue.

COMPLIANCE · **DPDP Act 2023**

Consent notice, purpose limitation, data-principal rights (access, correction, erasure, grievance), breach notification and retention limits apply to all member personal data. Legal review of the consent notice and grievance process is required before GA.

5. Non-Functional Requirements

ID	Category	Requirement	Target / Metric	Verification
NFR-01	Performance	The member app shall cold-start to an interactive Home screen quickly on a mid-range device.	≤ 3 s on Android, 4 GB RAM, 4G	Device-lab timing, 20 runs, P95
NFR-02	Performance	Read API endpoints shall respond within the target latency.	P95 ≤ 500 ms at 500 req/s	Load test (Locust/k6)
NFR-03	Performance	Write API endpoints shall respond within the target latency.	P95 ≤ 1 s at 500 req/s	Load test
NFR-04	Performance	Matrimony search with up to 4 filters shall return results end to end within the target.	≤ 2 s P95 over 50,000 profiles on 4G	Load test + device timing
NFR-05	Performance	News push notifications shall reach devices within the target after publish.	95 % of devices ≤ 30 s	Push delivery reports, staging test
NFR-06	Performance	Bulk import shall process large files within the target.	10,000 rows ≤ 2 min incl. dedup report	Timed test
NFR-07	Scalability	The system shall support concurrent users without degradation of NFR-02/03.	5,000 concurrent; 500 req/s peak; horizontal scaling of API instances	Soak test 2 h
NFR-08	Scalability	The data model shall support projected growth.	50,000 members across ≥ 50 cities with NFRs met	Load test on seeded volume
NFR-09	Availability	Production services shall meet the monthly uptime target.	99.5 % monthly, excluding announced 02:00 AM IST window	Uptime monitoring report
NFR-10	Reliability	Backups shall meet recovery objectives.	Daily automated PostgreSQL backup + PITR; full backup before every release; RPO ≤ 24 h; RTO ≤ 4 h	Quarterly restore drill
NFR-11	Reliability	Panchang shall be available daily before the cut-off.	Published before 05:00 AM IST on 100 % of days	Publish timestamp report
NFR-12	Reliability	The member app shall be stable.	Crash-free sessions ≥ 99.5 %	Crash reporting (e.g. Firebase Crashlytics)
NFR-13	Security	All PII shall be encrypted in transit.	TLS 1.2+ on all connections; HSTS on dashboard; no plaintext HTTP	TLS scan (SSL Labs grade A)

ID	Category	Requirement	Target / Metric	Verification
NFR-14	Security	All PII, ID documents and media shall be encrypted at rest; biodata fields shall be additionally encrypted at field level.	AES-256 storage encryption; field-level AES-256-GCM for biodata with keys in a managed KMS	Configuration audit, DB inspection
NFR-15	Security	Authorisation shall be enforced server-side on every request (role + city scope + biodata pair check).	0 critical/high findings in quarterly penetration test	External pen test (OWASP ASVS L2)
NFR-16	Security	Member sessions shall use signed short-lived access tokens with refresh tokens.	Access token ≤ 15 min; refresh token ≤ 30 days, rotated, revocable	Code review, auth tests
NFR-17	Security	Production database access shall be restricted.	Senior technical staff only, via VPN; every session logged	Access review, monthly
NFR-18	Security	API shall be protected against abuse.	Rate limit 60 req/min per member, 10 req/min on OTP endpoints per IP	Automated rate-limit tests
NFR-19	Security	Known vulnerabilities shall be patched.	Monthly security patching; critical CVEs ≤ 7 days	Dependency scan in CI
NFR-20	Compliance	Processing shall comply with India's DPDP Act 2023.	Consent recorded for 100 % of members; erasure ≤ 30 days; breach notification per Act	Legal review before GA; consent audit
NFR-21	Compliance	Audit logs shall be retained and protected.	Append-only; retained ≥ 24 months; reviewed monthly	Audit table permission check
NFR-22	Compliance	Data retention shall follow SOP §8 and Samaj bylaws.	Archive at 12 months inactive; purge at 24 months	Retention job report
NFR-23	Compliance	SMS templates shall comply with TRAI DLT rules.	100 % templates DLT-registered	Gateway configuration audit
NFR-24	Usability & Accessibility	Member app shall meet accessibility standards.	WCAG 2.1 AA; dynamic font scaling to 200 %; touch targets ≥ 44×44 pt	Accessibility audit (axe, TalkBack, VoiceOver)
NFR-25	Usability	UI shall render correctly on small, medium and large screens.	No clipping on 4.7", 6.1", 6.7" and 10" tablet reference devices	Visual regression tests

ID	Category	Requirement	Target / Metric	Verification
NFR-26	Compatibility	Supported platforms.	Android 9+; iOS 15+; Admin Dashboard on latest 2 versions of Chrome and Safari	Device/browser matrix test
NFR-27	Maintainability	Code quality and test coverage.	Backend unit test coverage ≥ 80 %; typed APIs with OpenAPI 3 spec auto-generated; DB migrations version-controlled	CI quality gate
NFR-28	Maintainability	Release discipline per SOP §10.	48 h code freeze; QA staging sign-off; pre-release backup; CI/CD deploy at 02:00 AM IST; smoke test	Release checklist audit
NFR-29	Observability	Monitoring and logging.	Centralised structured logs; real-time alerts on ERR_DB_CON and error rate > 2 %; daily log review	Alert fire drill
NFR-30	Maintenance	Periodic maintenance per SOP §11.	Weekly health check & cache clear; monthly DB index optimisation; quarterly pen test	Maintenance log

SYSTEM LIMIT · **Hard system limits**

- OTP: 6 digits, 5-minute validity, 3 attempts, 5 sends per number per hour.
- Images and ID documents: JPG/PNG (ID also PDF), ≤ 2 MB each; full-screen ad 1080×1920 px.
- Matrimony photos: maximum 5 per profile (assumption, OQ-12).
- Biodata requests: 1 open request per pair; 20 new requests per member per day; 30-day resend lock after decline.
- Import: CSV/XLSX, ≤ 10 MB and ≤ 50,000 rows per file.
- API rate limits: 60 requests/min per member; 10 requests/min per IP on OTP endpoints.
- Admin idle timeout 30 minutes; member access token 15 minutes, refresh token 30 days.

6. Data Requirements

6.1 Logical Data Entities

Entity	Key Attributes	Retention	Sensitivity
City	city_id, name, state, city_admin_id	Life of system	Low
AdminUser	admin_id, role (Super/City), city_id, email, password hash (Argon2id), TOTP secret (encrypted), status	Life of account + 24 months	High
Member	member_id, mobile (unique), name, gender, DOB, city_id, address, status (Pending/Active/Rejected/Archived), last_login_at	Archive 12 months inactive; purge 24 months	High
Household / FamilyMember	household_id, head_member_id, relative name, relationship, DOB, approval status	As Member	High
IdentityDocument	doc_id, member_id, storage key, mime, size, reviewed_by	Delete 90 days after approval decision (ASM)	High
Consent	member_id, notice_version, purpose, granted_at, withdrawn_at	Life of account + 24 months	Medium
MatrimonyProfile	profile_id, member_id, public summary (age, education, city, family background), managed_by, moderation status	As Member	Medium
Biodata	profile_id, detailed personal/family fields — field-level encrypted	As Member	High
MatrimonyPhoto	photo_id, profile_id, storage key, moderation status, reason	As Member	Medium
BiodataRequest	request_id, requester_id, recipient_id, status (Pending/Accepted/Declined/Withdrawn), created_at, responded_at	As Member	High
ContentItem	content_id, type (News/Ad), title, body, media key, start_at, end_at, city_id/all, status, created_by	24 months after expiry	Low
Panchang	date, location, tithi, nakshatra, yoga, source (API/Manual), edited_by, edited_at	Indefinite	Low
DeviceToken	member_id, platform, fcm_token, last_seen_at, valid	Until invalid or logout	Medium
ImportJob	job_id, file key, uploaded_by, counts (new/updated/conflict/error), report	12 months	Medium
ExportJob	job_id, filters, row_count, exported_by, created_at (no password stored)	24 months	Medium

Entity	Key Attributes	Retention	Sensitivity
AuditLog	log_id, actor, action, target, before/after, ip, timestamp — append-only	≥ 24 months	High
SecurityEvent	event_id, actor, endpoint, target, reason (e.g. biodata denied), timestamp	≥ 24 months	High
Feedback / Report	id, member_id, target, text, status	24 months	Medium

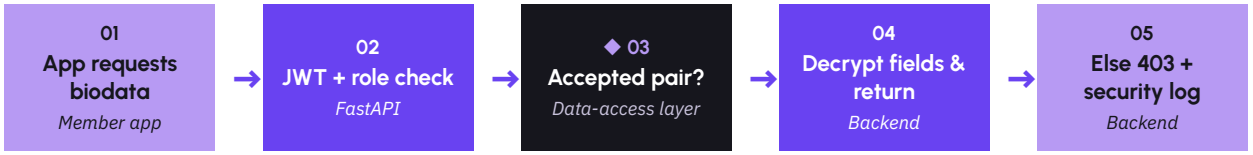
6.2 Data Quality Rules

- Mobile numbers stored in E.164 format (+91XXXXXXXXXX) with a unique constraint at database level.
- Dates stored as ISO 8601; timestamps in UTC (timestamptz) and displayed in IST.
- Import rows are validated per column type; invalid rows never reach the members table.
- Foreign keys and status enums are enforced at database level, not only in application code.
- Soft deletes use a deleted_at timestamp to support 30-day Archive Recovery.

6.3 Retention, Archival and Backup

- Inactive member profiles archived at 12 months and purged at 24 months (SOP §8); deleted profiles recoverable for 30 days (SOP §12).
- Audit and security logs retained at least 24 months and never purged by the member retention job.
- Daily automated PostgreSQL backups with point-in-time recovery; full database and file-storage backup before every release (SOP §10).
- Backups encrypted with AES-256 and stored in the same India region; restore tested quarterly.

Biodata data path



7. Requirements Traceability Matrix

Every SOP step is traced to features, functional requirements, related NFRs and test cases. Test case IDs are reserved for the QA test plan.

SOP Step	Feature	FR IDs	Related NFRs	Test Cases
§3 Roles; §4 Access & MFA	SF-11	FR-69–FR-74	NFR-15, NFR-16, NFR-17	TC-001–TC-006
§5.1 / §6.1 Steps 1–2 Registration, duplicate check	SF-01	FR-01–FR-08	NFR-02, NFR-13, NFR-16, NFR-18, NFR-20, NFR-23	TC-007–TC-014
§6.1 Steps 3–4 Moderator approval, activation push	SF-02	FR-09–FR-16	NFR-05, NFR-14, NFR-15	TC-015–TC-022
§3 Family additions	SF-03	FR-17–FR-18	NFR-15	TC-023–TC-024
§6.2 Listing & photo review	SF-04	FR-19–FR-24	NFR-14	TC-025–TC-030
§6.2 Browsing filters, 4 h index refresh	SF-05	FR-25–FR-28	NFR-04, NFR-08	TC-031–TC-034
§5.2 / §6.2 Steps 1–4 Biodata request & privacy rule	SF-06	FR-29–FR-42	NFR-14, NFR-15, NFR-21	TC-035–TC-048
§5.3 / §6.3 News & ads scheduling	SF-07	FR-43–FR-52	NFR-05	TC-049–TC-058
§9 Push ≤ 30 s; §12 FCM tokens	SF-08	FR-53–FR-56	NFR-05	TC-059–TC-062
§6.4 Panchang & manual override; §7 ERR_API_PN	SF-09	FR-57–FR-64	NFR-11, NFR-29	TC-063–TC-070
§2 / §3 Member directory	SF-10	FR-65–FR-68	NFR-02, NFR-15	TC-071–TC-074
§5.4 / §6.5 Import & deduplication	SF-12	FR-75–FR-81	NFR-06	TC-075–TC-081
§6.5 Encrypted filtered export	SF-13	FR-82–FR-85	NFR-14, NFR-21	TC-082–TC-085
§5.5 Monitoring; §7 Error codes; §8 Access logs	SF-14	FR-86–FR-91	NFR-21, NFR-29	TC-086–TC-091
§8 Retention; §12 Archive Recovery	SF-15	FR-92–FR-96	NFR-10, NFR-22	TC-092–TC-096
§8 Compliance (DPDP)	SF-16	FR-97–FR-100	NFR-20	TC-097–TC-100
§8 PII encryption	All	FR-20, FR-35	NFR-13, NFR-14	TC-SEC-01–TC-SEC-06

SOP Step	Feature	FR IDs	Related NFRs	Test Cases
§9 Staging test checklist	SF-01, SF-05, SF-06, SF-08, SF-12	FR-01, FR-25, FR-35, FR-54, FR-77	NFR-05, NFR-25	Release regression suite
§10 Deployment & release	—	—	NFR-10, NFR-28	Release checklist
§11 Maintenance & monitoring	SF-14	FR-90	NFR-29, NFR-30	Ops runbook checks

8. Appendix

8.1 Open Questions

ID	Question	Owner	Due
OQ-01	Are Moderator and Content Admin separate roles, or City Admin duties (ASM-02)?	Super Admin	Before design
OQ-02	Can any admin ever view biodata (e.g. abuse investigation)? Is withdrawal of an accepted exchange (FR-41) required?	Community leadership	Before design
OQ-03	Admin Dashboard technology: React Native for Web (shared code) or a separate React web app?	Vrattiks Architect	Before design
OQ-04	Which astrology API provider, licence and cost? Which reference location for sunrise-based Panchang?	Super Admin	Before Alpha
OQ-05	Which languages are required at launch (English, Hindi, Gujarati, others)?	Community leadership	Before design
OQ-06	Cloud host: AWS or Azure (SOP §7 mentions both), and which India region?	Vrattiks / Client	Before Alpha
OQ-07	Source and owner of weekly member lists and the import template?	City Admins	Before Beta
OQ-08	Which directory fields are visible to all members by default?	Community leadership	Before design
OQ-09	Confirm the daily biodata request cap (ASM-09) and 30-day resend lock (ASM-08).	Product	Before Beta
OQ-10	Retention period for identity documents after approval?	Data Protection Contact	Before Alpha
OQ-11	Background job stack (e.g. Celery + Redis, or APScheduler/ARQ) and hosting?	Vrattiks Architect	Before design
OQ-12	Are biodata fields fixed by the community, or configurable by the Super Admin?	Community leadership	Before design

8.2 Risks

ID	Risk	Impact	Likelihood	Mitigation
RSK-01	Biodata leaks to a third party through an API, export or admin view	High	Low	Central pair-access check in data-access layer; field-level encryption; privacy test suite as release gate; security logging
RSK-02	Astrology API outage near 05:00 AM	Medium	Medium	Evening pre-fetch, 04:00 alert, manual override

ID	Risk	Impact	Likelihood	Mitigation
RSK-03	Approval backlog because City Admins are volunteers	Medium	High	Queue ageing alerts, bulk approve, 72 h Super Admin escalation
RSK-04	Fake or impersonated matrimony profiles	High	Medium	Matrimony for verified members only; photo moderation; report-profile option
RSK-05	Poor-quality import files create bad data	Medium	High	Strict header validation, dry-run report, conflict review
RSK-06	Non-compliance with the DPDP Act	High	Low	Consent capture, in-app rights, retention automation, legal review before GA
RSK-07	SMS OTP cost and delivery failures	Medium	Medium	Rate limits, DLT templates, secondary SMS provider
RSK-08	Low adoption among older members	Medium	Medium	OTP-only login, large-text mode, regional languages, help at community events
RSK-09	Search performance degrades at 50,000 profiles on PostgreSQL	Medium	Low	Indexed filter columns, materialised search view refreshed ≤ 4 h, load test at target volume

Change control and sign-off

This SRS becomes the requirements baseline once approved by the signatories in Document Control. Any change to scope, priorities or requirement wording after sign-off must go through a change request that records impact on cost, schedule and the traceability matrix. Requirement IDs are never renumbered; retired requirements are marked as such.